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Consumer preferences for organically and locally produced apples

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This study investigates organic consumers' preferences for local production of apples. The analysis is based on a choice experiment among 637 Danish consumers used in combination with a principal component analysis of a set of opinion questions. The principal component analysis identifies two components of questions. Component 1 concerns benefits related to organic products while component 2 relates to positive features of locally produced products. When the components are included in analysis of data from the choice experiment a random parameter error component model suggests that respondents who recognize the benefits of organic products have relatively high preferences for both organically and locally produced apples. Respondents who, on the other hand, recognize the benefits of locally produced products have high preferences for domestically and locally produced apples but not for organic apples.

1. Introduction

Organic food sales are increasing around the world, particularly in the USA, Canada, and Europe. In Europe, the majority of the sales are occurring in the large economies (mainly Germany, France, the United Kingdom, and Italy), while the highest market share of organic food is found in Denmark followed by the alpine countries (Jensen, 2008; Willer, 2010).

There is a considerable literature which attempts to understand consumer perceptions of organic food and discusses the factors which determine organic food consumption (Aertens, Verbeke, Mondelaers, & Van Huylenbroeck, 2009). Socio-demographic factors, such as urbanization, income, education and gender have been identified as important correlates of organic consumption (Bellows, Onyango, Diamond, & Hallman, 2008; Jonas & Roosen, 2008; Monier et al., 2009), but there is also increasing acceptance that self-seeking interests are important when trying to understand the motives behind organic consumption (Aertens et al., 2009). Consumers with a high organic demand are more often convinced that organic products are healthier, better tasting, or more fresh than conventional products (Wier and Calverley, 2002). In addition to such private good attributes, organic food products also contain attributes which can be characterized as public goods, for example, lower use of pesticides and chemical fertilizers and improved conditions for animals involved in the production. A number of studies address the importance of such private and public good attributes in product perception or consumption (Aertens et al., 2009; Aldanondo-Ochoa & Almanza-Sáez, 2009; Fotopoulos, Krystallis, & Ness, 2003; Schifferstein & Ophuist, 1998), and several studies suggest that private good attributes, such as health and hedonic quality, contribute more to consumers' interest than public good attributes. On the other hand, the most often stated reason for not buying organic is the existence of considerable price premiums (Aertens et al., 2009; Gracia and de Magistris, 2008; Michaelidou & Hassan, 2010; Verhoef, 2005).

Today, there is an increasing trend that production of organic food can be characterized by large scale production, and large-scale retailers are becoming major players on the organic market in Denmark (Denver, Christensen, & Krarup, 2007), as well as in other countries with increasing demand for organic (Sahota, 2007). Large scale produced organic products have to fulfill the organic standards but may not necessarily be produced with additional principles, such as reduced food miles, environmentally friendly transportation or requests for fair trade. However, many consumers believe that the organic label is more comprehensive than it actually is and therefore attach such attributes to organic products (Andersen, 2009).

The international scientific literature in the field also suggests that many consumers are interested in locally produced food (see e.g. Carpio & Isengildina-Massa, 2009; Darby, Batte, Ernst, & Roe, 2008; Darby and Ernst, 2006; Giraud, Bond, & Bond, 2005; Hébert, 2011; Loureiro & Hine, 2002). There are various reasons for this positive consumer interest. Several studies indicate that local food is associated with higher perceived food quality (e.g. Carpio & Isengildina-Massa, 2009) as well as perceived increased freshness of the products (Darby et al., 2008; Roininen, Arvola, & Lähteenmäki,

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2006). Furthermore, short supply chains increase trustworthiness due to more direct communication between producers and consumers (Roininen et al., 2006).

Claims are also made that local food systems may have positive externalities, because they promote local jobs and help local business gaining market access (Carpio & Isengildina-Massa, 2009), and consumers may therefore feel that they support the local community when they purchase locally produced food (Bond, Thilmany, & Bond, 2008; Kuznesof, Tregear, and Moxey, 1997; Toler, Briggeman, Lusk, & Adams, 2009). In addition, local food production may also imply environmental benefits due to reduced 'food miles' (European Union – Committee of the Regions, 2011; Marsden & Smith, 2005).

Parts of the literature suggest that consumers may associate some of the same features with local food as with organic foods, for example environmental friendliness, animal welfare, and naturalness (Chambers, Lobb, Butler, Harvey, & Traill, 2007; Zepeda & Deal, 2009). But empirical studies also suggest some ambiguity with respect to the appeal of organic and local food to the consumers and, to the authors' knowledge, the interrelations between consumers' perceptions and demands with regard to organic and local foods have not been investigated quantitatively in the literature.

Against this backdrop, the main objective of the present study is to investigate the patterns in consumers' perception of organic and locally produced foods, respectively, and to examine the role of such perception in consumers' preferences for organic and local foods. This research objective is pursued by developing a methodology combining principal components analysis with econometric choice modeling as described in the next section. The methodology is demonstrated using Danish consumers' demand for organic and/or locally produced apples as a case. Apples is a group of products, where local conditions (climate, soil quality, etc.) influence the gastronomic quality, and in a Danish context, apples are often sold locally and/or as organic apples.

2. Methodology

The study addresses two aspects related to Danish consumers' demand for specific food varieties, such as organic or local products: the consumers' perception of the varieties, and their preference for the varieties, within a micro-economic utility framework. Based on an assumption of utility maximizing consumers, we consider the consumers' choice of apple varieties (organic, local, etc.) to depend positively on the consumers' perception of the different varieties, and negatively on the price. From this perspective, consumers' choice can be interpreted as a trade-off between product perception and price. This implies a higher relatively willingness to pay (WTP) for a certain variety (e.g. local), the more positively this variety is perceived, relative to the alternatives.

Product perception may be mediated by the consumers' values and norms, for example such that a positive perception of an organic variety's environmental impact will affect the consumption decision more strongly for a consumer who is concerned about the environment. Hence, some consumers may exhibit a high WTP for certain perceived characteristics, because they have strong opinions on these characteristics, and others have a lower WTP for these specific characteristics, but perhaps higher WTP for other perceived characteristics. One could imagine a continuous space of such values, norms and attitudes (perhaps correlated with observable values, such as age, family size, education, etc.) (e.g. Chamberlain, Kelley, & Hyde, 2013) or alternatively a segmentation, where consumers fall in different clusters with emphasis on certain sets of values, etc (Baker and Burnham, 2001).

2.1. Data

Data for the study were generated through an internet[2] questionnaire survey in a sample from Userneeds'[3] Danish online panel database in 2010. Panel members are recruited for market surveying by Userneeds in an attempt to obtain a representative sample of Danish consumers in the age range 18–65 years, in terms of gender, age, family structure, income and region. Hence, although apples are consumed on a regular basis by a vast majority of Danes, there were respondents who did not consume apples (about 5%).

The questionnaire was developed on the basis of insights from the above-mentioned literature and incorporated statements regarding respondents' perception of organic and locally produced apples (e.g. Roininen et al., 2006; Verhoef, 2005). Respondents were asked to declare their level of agreement with these statements (for example, "Organic products are more environmentally friendly") on a 5-point scale (fully disagree, partially disagree, neither agree nor disagree, partially agree, fully agree) – with the possibility to answer "don't know". The set of replies to statements reflecting perception of organic and locally produced food amounted to 14 variables, which could be expected to have an impact on consumers' preferences and hence their propensity to choose organic and/or local product varieties. The questionnaire was pilot tested on a smaller sample with 104 individuals prior to the final survey.

In order to address the respondents' propensity to choose organic and/or local product varieties, the questionnaire also included a hypothetical choice experiment, where participants were requested to make choices between different combinations of attributes in apples, assuming that these choices reflected their preferences for these combinations in a sequence of choice sessions. In each choice session, the respondent was asked to choose between three alternative apple varieties, of which "Apple A" and "Apple B" represented different levels of the attributes listed in Table 1, and "Apple C" represented a "status quo" reference (conventionally grown, imported from outside EU, mixed colour, sour taste and mealy texture at a price of 7 DKK/kg). The list of attributes beyond production method (organic, conventional) and geographic origin (local, Denmark, other EU country or outside EU), as well as the levels for these attributes, were determined on the basis of insights from Hampson et al. (2000). By varying the composition of products' attributes (e.g. with respect to type of production, region of origin, sensory attributes etc.) in the hypothetical choice settings, it is thus possible to generate data on the respondents' preferences for the different attributes in the apples.

In order to maximize the amount of information about respondents' preferences from the choice experiments, a D-efficient Bayesian updated fractional factorial experimental design resulting in 12 different choice sets in total was applied, using the software Ngene (Rose et al., 2009). The updated design was based on results from the pilot study with 104 respondents.

2.2. Statistical analysis

As some of the 14 variables reflecting perceptions of organic and locally produced apples were suspected to be mutually correlated, principal component analysis was used to aggregate these variables into a smaller number of orthogonal factors as a first step in the statistical analysis. Principal components analysis enables the identification of more general structures in consumers' product perception, and furthermore eliminates potential problems of multicollinearity. In particular, the number of components to be

[2] In 2010, 86 % of Danish households had access to the internet in their homes (Statistics Denmark, 2010).

[3] http://www.userneeds.dk.

Table 1
Apple attributes and their levels in the hypothetical choice experiment.

Characteristics	Levels
Type of production	Conventional, organic
Origin	Local, Danish, other EU country, outside EU
Colour of apples	Red, green, yellow, mix of colours
Taste and texture	Sweet and crunchy, sweet and mealy, sour and crunchy, sour and mealy
Price (DKK/kg)*	7, 15, 25, 45

* 1 DKK–0.13 €.

Table 2
Socio-demographic description of the respondents.

	Panel (%)	Adult citizens of Denmark (%)
Gender		
Male	306 (48.1)	50.3
Female	330 (51.9)	49.7
Age in years		
19–34	229 (36.0)	29.6
35–49	209 (32.9)	35.9
50–65	198 (31.1)	34.5
Education		
Maximum high-school	176 (27.6)	33.3
Professional	341 (53.5)	58.4
Academic	121 (19.0)	8.3
Place of living		
Copenhagen	210 (33.0)	31.7
Outside Copenhagen	427 (67.0)	68.3
Total respondents	637 (100)	

retained was determined as the number of eigen-values higher than one in the matrix consisting of the 14 variables. Each of these factors was given an interpretation on the basis of varimax-rotated factor loadings. This was done using proc factor in SAS 9.2®.

The choice experiment includes the choice set $i = \{AppleA, AppleB, AppleC\}$ and the choices are assumed to be driven by utility maximization behaviour. The utility function for respondent n is assumed to be described as $U_{nj} = V_{nj} + \varepsilon_{nj}$, where V_{nj} is the systematic (observed) part of the utility and $\varepsilon_{nj} \sim iid$ is a random (unobservable) part of the utility (Train, 2009). The observed part of the utility is assumed to be a linear function of the observed attributes in the choice experiment. In order to capture the relationship between the respondents' perception and their preferences for the non-price attributes of apples, each attribute was interacted with the respondent's scores on the principal factors, that is, $V_{nj} = \boldsymbol{\alpha}'_n\mathbf{x}_{nj} + \beta_n\mathbf{z}_{nj}\mathbf{x}'_{nj} + \gamma_n p_j$ where the first term on the right-hand side, α'_n, captures the vector of utilities related to the four non-price product attributes and the second term, β_n, captures the four non-price attributes interacted with observed scores from the PCA of the respondents, $\mathbf{z}_{nj}$. The last term on the right-hand side captures the weight, γ_n, put on the price attribute, p_j. Alternative specific constants, Asc_j, were included to capture preferences for unobserved attributes of apples.

Instead of including socio-demographic information about the consumers, a random parameter error component model was used in order to account for taste heterogeneity among the respondents (Greene & Hensher, 2007). In the estimation, all non-monetary attribute parameters were assumed to be normally distributed. Hereby, we allow for both negative and positive preferences for the attributes and estimate means and standard deviations. The price parameter is, on the other hand, kept fixed.

A stronger correlation between the first two alternatives (*Apple A* and *Apple B*) than between these two alternatives and the third "status quo" alternative (*Apple C*) was expected. In order to capture this asymmetry in correlation, we introduced the term $Apple_j\,\delta$, in which δ was a normally distributed error component with mean zero, and $Apple_j$ was a dummy that took the value 1 if alternative j was one of the first two variants of apples and 0 if the third alternative was chosen.

A main effects model was used in order to capture the marginal effects of the apples' attributes on the probability of choosing a given bag of apples. The econometric estimation was performed by BIOGEME 1.8 (Bierlaire, 2003) using 300 pseudo-random draws in simulating the random parameter distributions.

3. Results

A total of 637 respondents completed the questionnaire. The socio-demographic distributions with respect to gender, age, education, and place of living of the samples are presented in Table 2 and compared with the corresponding distributions of the Danish population.

Women, younger people, respondents with a higher education, and inhabitants of Copenhagen are slightly overrepresented compared to the Danish population (Statistics Denmark, 2011). This is in line with earlier studies which show that online panels typically include too few people of the older generation and too many highly educated (Olsen, 2009).

Regarding normal choices of apple varieties, 2% of the respondents stated that they normally buy apples that are organic and local, 24% stated that they buy organic but not local apples, and 3% stated to normally buy local, but not organic, apples.

3.1. Results of principal factor analysis

The pool of 14 variables analyzed in the principal factor analysis resulted in two principal factors. Factor 1 (C1) had an eigenvalue of 4.15 while Factor 2 (C2) had an eigenvalue of 1.69. Kaiser's Measure of Sampling Adequacy was 0.85 and the two factors explained approximately 50% of the common variation in the data. Rotating the factors using varimax rotation, they were found to represent different dimensions in respondents' statements. As shown in Table 3, variables concerning perceived benefits of organic products had the largest loadings on C1, and variables concerning benefits of local production had lower, but still positive, loadings on this first component. Contrary to this, variables concerning organic production had lower loadings on C2, while variables concerning local production had higher loadings in this factor.

Further analyses (not presented) indicated that positive perception of organic apple varieties was associated with gender and geography, in that women and respondents in eastern Denmark tended to have above-average perceptions of organic apples, but no statistically significant association was found with other socio-demographic variables, such as age, education and income. No significant correlations were found between perception of local varieties of apples and socio-demographic variables. Age was found to display a weak (but not statistically significant) positive association with perception of local apples and a weak (and also not significant) negative association with the respondents' perception of organic apples.

3.2. Results of choice experiment

Results from econometric analysis of consumer preferences as reflected in the choice experiment data are shown in Table 4. The reference apple used in the estimation is a conventional apple with a mix of colours produced outside the EU having a sour taste and mealy texture, and the presented results represent deviations from

Table 3
Product perception components – varimax rotated factors from PCA.

	C1	C2
Share of total variation explained	0.34	0.16
Organic products are more environmentally friendly	0.65	0.05
Organic products are healthier	0.81	0.11
Organic products taste better	0.77	0.21
There are no pesticide residues in organic products	0.53	0.06
Organic products look more appealing	0.57	0.28
The quality is better in organic products	0.75	0.25
Organic products are easier to find in the shop	0.15	0.21
Organic products are cheaper	0.10	0.29
Locally produced goods are more environmentally friendly	0.23	0.46
Locally produced goods are healthier	0.24	0.74
Locally produced goods taste better	0.13	0.83
The quality is better in locally produced goods	0.12	0.78
Locally produced goods are easier to find in the shop	0.00	0.50
Locally produced goods are cheaper	0.05	0.44

Table 4
Estimation results for random parameter error component model for utility derived from apple attributes.

Variable	Coefficient	Robust standard error	Standard deviation	Robust standard error
ASC_A	1.62***	(0.13)		
ASC_B	1.61***	(0.13)		
δ	2.02***	(0.11)		
Local	1.90***	(0.09)	0.35	(0.66)
Local*C1	0.36***	(0.10)	0.14	(0.55)
Local*C2	0.33***	(0.08)	0.23	(0.38)
Danish	1.63***	(0.09)	0.05	(0.19)
Danish*C1	0.20*	(0.09)	0.07	(0.19)
Danish*C2	0.44***	(0.08)	0.20	(0.37)
EU	0.66***	(0.07)	0.70*	(0.30)
EU*C1	0.22***	(0.06)	0.09	(0.31)
EU*C2	−0.01	(0.07)	0.20	(0.36)
Organic	0.54***	(0.06)	0.13	(0.29)
Organic*C1	0.68***	(0.08)	0.65	(0.36)
Organic*C2	0.03	(0.06)	0.04	(0.32)
Red	0.15*	(0.07)	0.05	(0.17)
Green	−0.25***	(0.07)	1.32	(0.70)
Yellow	−0.34***	(0.07)	0.31	(0.45)
Sweet/mealy	0.14	(0.11)	0.53**	(0.19)
Sweet/crunchy	3.13***	(0.15)	0.25	(0.39)
Sour/crunchy	2.61***	(0.12)	0.58	(0.37)
Price	−0.10***	(0.00)		
Adj. ρ^2	0.23			

* Indicate significance at 0.05 level, respectively.
** Indicate significance at 0.01 level, respectively.
*** Indicate significance at 0.001 level, respectively.

this reference apple type. Robust (i.e. heteroschedasticity-adjusted) standard errors were estimated for both the preference coefficients and their standard deviations, in order to assess the statistical significance of these parameters. All coefficients, except EU*C2, Organic*C2 and Sweet/mealy were significant, whereas only two standard deviations (EU and Sweet/mealy) were statistically significant. For example, the estimation suggests that rather than apples produced in countries outside the EU, respondents prefer locally produced apples, domestically produced apples and apples coming from other EU-countries (in that order). A significant standard deviation for apples produced in other EU-countries can be interpreted as some spreading in preferences concerning this attribute.

Furthermore, there is a positive relationship between both the organic and the local perception components from the principal components analysis and the estimated preferences for local and domestic apples. In particular, a positive perception of either organic or local apples is positively correlated with high preferences for local apples, while Danish apples are mainly preferred by respondents with positive perception of the features of local apples.

Respondents in general prefer organic apples to non-organic apples, and respondents scoring high on the organic component from the principal component analysis also have relatively high preferences for organic apples. On the other hand, there is no significant relationship between the local component and tendency to select organic apples.

With respect to the color of the apples, the respondents have relatively low preferences for green and yellow apples but high preferences for red apples relative to apples with more than one color. Very high coefficients for crunchy apples stress that the taste and texture of the apples is of major importance for the respondents. A significant standard deviation for sweet and mealy apples suggests that some respondents may prefer these instead of sour and mealy apples, but a considerable share of the respondents have the opposite taste ordering of mealy apples. The price parameter is negative, which suggests disutility of higher prices as expected. The significant error component δ shows that there is correlation between choices made by the same respondent.

The coefficient estimates in Table 4 can be transformed into willingness-to-pay estimates for the different non-price attributes as the negative ratio between the coefficient associated with the attribute and the price coefficient (Hu, Batte, Woods, & Ernst, 2011). For example, this implies that the respondents on average are willing to pay a price premium of 5.40 DKK/kg for organic apples and a price premium of 19.00 DKK/kg for locally produced apples – even if they have a neutral perception of organic apples (as represented by the factor C1, cf. Table 3), or locally produced apples (factor C2). And if they have maximum perception of the organic attributes, they are willing to pay a price premium of 12.20 DKK/kg for organic apples and 22.60 DKK/kg for locally produced apples.

4. Discussion

The two principal components in respondents' perception of organic or local apples suggest that the two product types tend to appeal to different consumers, with some consumers having high expectations regarding organic apples, but these are not necessarily the same as those consumers with high perception of local products. The choice experiment analysis suggests that consumers with positive perception of organic apples also have relatively strong preference for apples being locally produced, but respondents with high perception of local products did not have stronger preferences for the organic variety relative to other consumers. This asymmetry is interesting and might suggest new, and relatively unexplored, market potentials, for example a potential high-end market for organic and locally produced foods. Further research is however needed to substantiate this interpretation.

Whereas the modeling of interaction between consumers' perceptions and preference parameters in willingness-to-pay analysis to the authors' knowledge is new, parts of the above findings are in line with findings from other studies in the literature. In particular, Hu et al. (2011) conducted a choice experiment in Ohio and Kentucky, USA, concerning blackberry jam including different levels of organic production and places of production, with results suggesting that respondents were more likely to select organic products and products produced within their own state. On the other hand, Onozaka and McFadden (2011) studied U.S. consumer preferences for tomatoes and apples, and did not find any significant interaction effect between preferences for organic and local production, and Yue and Tong (2009) even found a negative interaction effect between organic and locally produced tomatoes in Minnesota.

Results of the principal components analysis suggest that consumers' perception of 'organic' and 'local' attributes constitute different dimensions with different scores on the two dimensions for different consumers. Whereas our results regarding consumers' perception of these attributes did only show weak associations with socio-demographic variables, Chamberlain et al. (2013) found that age tended to have a stronger positive effect on consumers' preference for local products than for organic products. Studies by Bean and Sharp (2011) and Pouta, Heikkilä, Forsman-Hugg, Isoniemi, and Mäkelä (2010) found different segments of consumers, with some emphasizing organic attributes, some emphasizing geographic origin, and some segments either modest or very inclined to both organic and local food. On the other hand, Nie and Zepeda (2011) found that 60% of the consumers were either very or modestly interested in both organically and locally produced food, while the last 40% were uninterested in both types of food. Results from a contingent valuation study by Loureiro and Hine (2002) show that consumers in Colorado, USA, are willing to pay a higher price premium for Colorado grown potatoes than for organic potatoes. Zepeda and Deal (2009) reached a similar conclusion, and this also tends to be in line with findings in our study, where the estimated coefficient for the 'local' attribute is significantly larger than that of the 'organic' attribute.

Results from our study are hence in line with findings from a range of other studies with different research angles, methodologies and perspectives, and focusing on either consumers' perceptions or attitudes, or on their choices and willingness to pay,with regards to organic or locally produced foods. Our study illuminates the interrelations between product perceptions and stated behaviour by bringing these different aspects together in one quantitative analytical framework, with results that make sense in light of previous findings.

The identified interaction effects suggest that measures to enhance consumers' perception of organic or local apples may also have stimulating effects on the demand for some of the other varieties. For example, by defining higher standards for local product varieties – and enhancing consumers' knowledge and perception of products fulfilling these standards – the demand for domestic produce may also be enhanced. Or by raising the standards – and communication of these standards – for organic products, the demand and willingness to pay for local products may be enhanced as well, however probably depending on the specific nature of such standards. As a consequence, authorities could include requirements to place of production as a part of an extended organic standard. By introducing such an 'Organic+' label, the integrity of organic products among more dedicated organic consumers – and consumers with preference for local products – might increase, while (less dedicated) consumers primarily interested in basic organic products could be maintained. However, more research is necessary in order to interpret these interaction effects in more detail.

Naturally, the robustness of the results can be questioned as they are based on a single product. With regard to local products in Denmark, apples may constitute a special case, due to a relatively widespread availability of apples from both commercial and non-commercial outlets in the season. An extension of the analysis to include other food products therefore immediately springs to mind as an obvious way to further validate the present findings and increase their robustness.

All questions included in the principal components analysis regarding local products concern perceived quality of these local products, while support to local community through purchases of locally produced products was not addressed in the questionnaire. As some previous studies (e.g. Caprio & Isengildina-Massa, 2009; Marsden, Banks, & Bristow, 2000) have pointed out local employment concerns as a potential driver for consumption of local products, inclusion of such aspects might have yielded different results, depending on, whether and how consumers' concern for such issues are correlated with aspects covered in C1 and C2 components.

Whereas products eligible for being labeled as 'organic' must fulfill a list of specific requirements, there are no such requirements to 'local' products. Hence, the consumers' perception of characteristics related to organic apples may be much more substantiated and homogenous than their perception of local apples' characteristics (which may also have been reflected in the larger estimated standard deviations associated with preference parameters related to geographic origin and factor C2 than to the parameters related to C1 and organic attributes).

Furthermore, it should also be taken into account that the respondents in the panel are better educated and more likely to live in the capital area than the average Danish consumer. As outlined in the introduction, these socio-demographic characteristics also apply to organic consumers in general. The data are therefore very suitable to describe if there is a demand for local and Danish produced apples among organic consumers (but perhaps slightly less representative with respect to non-organic consumers). As the importance of organic production can thus be somewhat overstated, some precautions need to be taken in order to extend the results to the Danish population in general.

5. Conclusion

In this questionnaire-based study of Danish apple consumers, we found two principal components in respondents' stated perceptions of organically and locally produced apples. Whereas the first component is correlated with respondents' preference for organic apples, the second component is related to their preference for locally produced apples. But important cross effects were also identified, in that positive perception of organic products affects the preference for locally produced apples positively.

Our results stress that the taste of the apples is the most important attribute for the consumers, but that there is an interest for both local and domestic production among the consumers. The results also indicate that consumers are willing to pay an additional price premium for both Danish and local production, and this willingness to pay is enhanced, if the consumers have stronger positive perception of the organic or local apples, respectively. One interesting feature of the results is the apparent existence of interaction effects between consumers' perception of organic apples and their willingness to pay for local apples, and between their perception of local apples and their willingness to pay for domestic – but not necessarily local – apples. Such cross-effects might carry new – and so far under-exploited – market potentials, but further research is needed in order to clarify such potentials.

References

Aertens, J., Verbeke, W., Mondelaers, K., & Van Huylenbroeck, G. (2009). Personal determinants of organic food consumption: A review. *British Food Journal, 111*(10), 1140–1167.

Aldonondo-Ochoa, A. M., & Almanza-Sáez, C. (2009). The private provision of public environment: Consumer preferences for organic production systems. *Land Use Policy, 26*(3), 669–682.

Andersen, L. M. (2001). *Documentation of CONCEPTS questionnaires*. Institute of Food and Resource Economics, University of Copenhagen.

Baker, G. A., & Burnham, T. A. (2001). Consumer response to genetically modified foods: Market segment analysis and implications for producers and policy makers. *Journal of Agricultural and Resource Economics, 26*(2), 387–403.

Bean, M., & Sharp, J. S. (2011). Profiling alternative food system supporters: The personal and social basis of local and organic food support. *Renewable Agriculture and Food Systems, 26*(3), 243–254.

Bellows, A. C., Onyango, B., Diamond, A., & Hallman, W. K. (2008). Understanding consumer interest in organics: Production values vs. purchasing behaviour.

Journal of Agricultural and Food Industrial Organization, 6, article 2. <http://www.degruyter.com/view/j/jafio.2008.6.1/jafio.2008.6.1.1169/jafio.2008.6.1.1169.xml> Accessed 20.09.2012.

Bierlaire, M. (2003). *BIOGEME: A free package for the estimation of discrete choice models", proceedings of the 3rd Swiss transportation research conference*. Switzerland: Ascona.

Bond, C. A., Thilmany, D., & Bond, J. K. (2008). Understanding consumer interest in product and process-based attributes for fresh produce. *Agribusiness, 24*(2), 231–252.

Carpio, C. E., & Isengildina-Massa, O. (2009). Consumer willingness to pay for locally grown products: The case of South Carolina. *Agribusiness, 25*(3), 412–426.

Chamberlain, A. J., Kelley, K. M., & Hyde, J. (2013). Locally grown and certified organic produce in the mid-atlantic region of the United States. *HortTechnology, 23*(1), 74–81.

Chambers, S., Lobb, A., Butler, L., Harvey, K., & Traill, W. B. (2007). Local, national and imported foods: A qualitative study. *Appetite, 49*(1), 208–213.

Darby, K., Batte, M. T., Ernst, S., & Roe, B. (2008). Decomposing local: A conjoint analysis of locally produced foods. *American Journal of Agricultural Economics, 90*(2), 476–486.

Darby K., Batte M.T., Ernst S. & Roe B. (2006) *Willingness to pay for locally produced foods: A customer intercept study of direct market and grocery store shopp*ers, paper presented at American Agricultural Economics Association Annual Meeting, Long Beach, California, July 23–26, 2006.

Denver, S., Christensen, T., & Krarup, S. (2007). Forbruget af økologiske fødevarer og ernæringsrigtig kost. *Samfundsøkonomen, 5*, 29–33.

European Union – Committee of the Regions (2011). Local food systems can boost rural development and improve the CAP, Committee of the Regions tells Commissioner Cioloş, press release COR/11/3, Brussels, January 27, 2011 <http://europa.eu/rapid/pressReleasesAction.do?reference=COR/11/3&format=PDF&aged=1&language=EN&guiLanguage=en>.

Fotopoulos, C., Krystallis, A., & Ness, M. (2003). Wine produced by organic grapes in Greece: Using means-end chains analysis to reveal organic buyers' purchasing motives in comparison to the non-buyers. *Food Quality and Preference, 14*(7), 549–566.

Giraud, K. L., Bond, C. A., & Bond, J. J. (2005). Consumer preferences for locally made specialty food products across northern New England. *Agricultural and Resource Economics Review, 34*(2), 204–216.

Gracia, A., & de Magistris, T. (2008). The demand for organic foods in the South of Italy: A discrete choice model. *Food Policy, 33*(5), 386–396.

Greene, W. H., & Hensher, D. A. (2007). Heteroscedastic control for random coefficients and error components in mixed logit. *Transportation research part E: Logistics and transportation review, 43*(5), 610–623.

Hampson, C. R., Quamme, H. A., Hall, J. W., MacDonald, R. A., King, M. C., & Cliff, M. A. (2000). Sensory evaluation as a selection tool in apple breeding. *Euphytica, 111*(2), 79–90.

Hébert, M. (2011). Examining current research on local food: A review. *Studies by Undergraduate Researchers at Guelph, 4*(2), 88–92.

Hu, W., Batte, M. T., Woods, T., & Ernst, S. (2011). Consumer preferences for local production and other value-added label claims for a processed food product. *European Review of Agricultural Economics, 39*(3), 489–510.

Jensen, K. L. (2008). Den økologiske markedsudvikling. In H. F. Alrøe & N. Halberg (Eds.), *Udvikling, vækst og integritet i den danske økologisektor: vidensyntese om muligheder og barrierer for fortsat udvikling og markedsbaseret vækst i produktion, forarbejdning og omsætning af økologiske produkter*. ICROFS: Foulum.

Jonas, A., & Roosen, J. (2008). Demand for milk labels in germany: Organic milk, conventional brands, and retail labels. *Agribusiness, 24*(2), 192–206.

Kuznetsof, S., Tregear, A., & Moxey, A. (1997). Regional foods: A consumer perspective. *British Food Journal, 99*(6), 199–2006.

Loureiro, M. L., & Hine (2002). Discovering niche markets: A comparison of consumer willingness to pay for local (Colorado grown), organic and GMO-free products. *Journal of Agricultural and Applied Economics, 34*(3), 477–487.

Marsden, T., Banks, J., & Bristow, G. (2000). Food supply chain approaches: Exploring their role in rural development. *Sociologia Ruralis, 40*(4), 424–438.

Marsden, T., & Smith, E. (2005). Ecological entrepeneurship: Sustainable development in local communicites through quality food production and local branding. *Geoforum, 36*(4), 440–451.

Michaelidou, N., & Hassan, L. M. (2010). Modeling the factors affecting rural consumers' purchase of organic and free-range produce: A case study of consumers' from the Island of Arran in Scotland, UK. *Food Policy, 35*(2), 130–139.

Monier S., Hassan D., Nichéle V. & Simioni M. (2009). *Organic food consumption patterns, Journal of Agricultural & Food Industrial Organization*, vol. 7, article 12, special issue: Quality promotion through eco-labeling.

Nie, C., & Zepeda, L. (2011). Lifestyle segmentation of US shoppers to examine organic and local food consumption. *Appetite, 57*, 28–37.

Olsen, S. B. (2009). Choosing between internet and mail survey modes for choice experiment surveys considering non-market goods. *Environmental and Resource Economics, 44*, 591–610.

Onozaka, Y., & McFadden, D. T. (2011). Does local labeling complement or compete with other sustainable labels? A conjoint analysis of direct and joint values for fresh produce claims. *American Journal of Agricultural Economics, 93*(3), 693–706.

Pouta, E., Heikkilä, J., Forsman-Hugg, S., Isoniemi, M., & Mäkelä, J. (2010). Consumer choice of broiler meat: The effects of country of origin and production methods. *Food Quality and Preferences, 21*, 539–546.

Roininen, K., Arvola, A., & Lähteenmäki, L. (2006). Exploring consumers' perceptions of local food with two different qualitative techniques: Laddering and word association. *Food Quality and Preference, 17*, 20–30.

Rose, J.M., Collins, A.T., Bliemer, M.C., Hensher, D.A. (2009). *Ngene 1.0 stated choice experiment design software*, University of Sydney.

Sahota A. (2007). Overview of the global market for organic food and drink, chapter 7 in Willer H. & Yussefi M.(Eds.) (2007) The World of Organic Agriculture. Statistics and Emerging Trends 2007. 9th edition, totally revised and updated, International Federation of Organic Agriculture Movements IFOAM, Bonn, Germany & Research Institute of Organic Agriculture FiBL, Frick, Switzerland, ISBN IFOAM: 3-934055-82-6, ISBN FiBL 978-03736-001-9©.

Schifferstein, H. J. N., & Ophuist, P. A. M. O. (1998). Health-related determinants of organic food consumption in the Netherlands. *Food Quality and Preference, 8*(3), 119–133.

Statistics Denmark (2010). *Befolkningens brug af internet – 2010. Statistics Denmark*. Available online: <http://www.dst.dk/pukora/epub/upload/15239/it.pdf>.

Statistics Denmark (2011). *Statistical Yearbook 2011. Danmarks Statistik, Denmark*. Available online: <http://www.dst.dk/pukora/epub/upload/15239/it.pdf>.

Toler, S., Briggeman, B. C., Lusk, J. L., & Adams, D. C. (2009). Fairness, farmers markets and local production. *American Journal of Agricultural Economics, 91*(5), 1273–1278.

Train, K. (2009). *Discrete choice methods with simulation* (2nd ed.). Cambridge University Press.

Verhoef, P. C. (2005). Explaining purchases of organic meat. *European Review of Agricultural Economics, 32*(2), 245–267.

Wier, M., & Calverley, C. (2002). Market potential for organic foods in Europe. *British Food Journal, 104*(1), 45–62.

Willer, H. (2010). Organic farming in Europe – An overview. In H. Willer & L. Kilcher (Eds.), *The world of organic agriculture, Statistics and emerging trends 2010*. IFOAM, Bonn and FiBL, Frick.

Yue, C., & Tong, C. (2009). Organic or local? Investigating consumer preference for fresh produce using a choice experiment with real economic incentives. *HortScience, 44*(2), 366–371.

Zepeda, L., & Deal, D. (2009). Organic and local food consumer behaviour: Alphabet theory. *International Journal of Consumer Studies, 33*(6), 697–705.